THE SMASH CONSTRUCTION FACTORS A MORPHISM THROUGH ONE ETALE OUTSIDE T AND ISOMORPHIC ON COMPLETIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file restricted.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the source proves the lemma in full, including the etale equivalence relation check and the completion isomorphism

Editorial note. The label lemma-smash-away-from-T names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- formal completion along a closed subset and the completion functor of the chapter
- glueing of algebraic spaces along open immersions to produce pushout spaces
- etale equivalence relations and the bootstrap quotient theorem
- checking finite type and isomorphism conditions etale locally

Source reference. Original theorem 04S6: sources/stacks-chow/tags.txt, source lines 6209–6209. Prerequisite pointer invoked at proof line(s) 6595; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0AIX: sources/stacks-chow/tags.txt, source lines 13280–13280. Prerequisite pointer invoked at proof line(s) 6607; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0GDB: sources/stacks-restricted/pinned-original-source.tex, source lines 5953–5953. Prerequisite pointer invoked at proof line(s) 6572; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02WR: sources/stacks-chow/tags.txt, source lines 3814–3814. Prerequisite pointer invoked at proof line(s) 6589; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:restricted.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let X be a locally Noetherian algebraic space over S and let $T \subset |X|$ be a closed subset. Let $f : X' \to X$ be a morphism of algebraic spaces which is locally of finite type and étale outside of T. There exists a factorization*

$$X' \longrightarrow X'' \longrightarrow X$$

of f with the following properties: $X'' \to X$ is locally of finite type, $X'' \to X$ is an isomorphism over $X \setminus T$, and $X'_{/T} \to X''_{/T}$ is an isomorphism (see proof for notation).

Proof. The notation using the subscript $_{/T}$ in the statement refers to the construction which to a morphism $f : X' \to X$ of algebraic spaces associates the morphism $f_{/T} : X'_{/f^{-1}T} \to X_{/T}$ of formal algebraic spaces, see Section 0GDB. We will also use the notion $U \subset X$ and $U' \subset X'$ to denote the open subspaces with $|U| = |X| \setminus T$ and $U' = |X'| \setminus f^{-1}T$ introduced in Section 0GDB.

After replacing X' by $X' \amalg U$ we may and do assume the image of $X' \to X$ contains U. Let

$$R = X' \amalg_{U'} (U' \times_U U')$$

be the pushout of $U' \to X'$ and the diagonal morphism $U' \to U' \times_U U' = U' \times_X U'$. Since $U' \to X$ is étale, this diagonal is an open immersion and we see that R is an algebraic space (this follows for example from Spaces, Lemma 02WR). The two projections $U' \times_U U' \to U'$ extend to R and we obtain two étale morphisms $s, t : R \to X'$. Checking on each piece separately we find that R is an étale equivalence relation on X'. Set $X'' = X'/R$ which is an algebraic space by Bootstrap, Theorem 04S6. By construction have the factorization as in the lemma and the morphism $X'' \to X$ is locally of finite type (as this can be checked étale locally, i.e., on X'). Since $U' \to U$ is a surjective étale morphism and since $s^{-1}(U') = t^{-1}(U') = U' \times_U U'$ we see that $U'' = U \times_X X'' \to U$ is an isomorphism. Finally, we have to show the morphism $X' \to X''$ induces an isomorphism $X'_{/T} \to X''_{/T}$. To see this, note that the formal completion of R along the inverse image of T is equal to the formal completion of X' along the inverse image of T by our choice of R! By our construction of the formal completion in Formal Spaces, Section 0AIX we have $X''_{/T} = (X'_{/T})/(R_{/T})$ as sheaves. Since $X'_{/T} = R_{/T}$ we conclude that $X'_{/T} = X''_{/T}$ and this finishes the proof. □